

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering

ASSESSMENT TOOL 2 – Debugging & Optimization Task

Course Code:	CSA0309	Course Title:	Data Structures
CO Assessed:	CO2 – Manipulation (BL3, BL4)	Assessment:	Assessment Tool 2 – Debugging & Optimisation Task
Weightage:	35% of CIA	Total Marks:	100
CO2 Statement:	Implement linear data structures such as stacks, queues, and linked lists, and analyze the efficiency of linear and binary search techniques.		
Student Name:	Pawan.C	Reg. No.:	192521202
Section / Batch:		Date:	22/07/2026

Code of Conduct

Pawan.C / 192521202

I _____ (Name / Reg No)

certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____

Pawan.C

Instructions

- This test contains 20 Debugging & Optimisation Task questions. Total: 100 Marks (20*5).
- When a student answer using Scenario-Based, in evaluation the answer should be with following five requirements: Identify the problem type and constraints, Select appropriate data structure, Design the Solution, Optimization and efficiency, Clarity of representation.
- No negative marking. Unanswered questions carry zero marks.

Section / Topic	Focus	Q Nos	Marks
Search Data Structure, Stack	Binary Search, Stack Operations	1 - 5	25
Linked List Data Structure	List Operations	6 - 10	25
Queue Data Structures	Queue Operations	11 - 15	25
Queue Data Structures, Stack Notations	Queue Operations, Expression Evaluation	16 - 20	25
GRAND TOTAL:			100 Marks

CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1)

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QUESTIONS

Q1 Binary Search
5 marks

Q2 Stack
5 marks

Q3 Stack
5 marks

Q4 Stack
5 marks

Q5 Stack
5 marks

Q6 Linked List
5 marks

Q7 Stack
5 marks

Q8 Linked List Data Structures
5 marks

Q9 Linked List
5 marks

Q10 Queue
5 marks

10/10 answered

Q1 Binary Search

5 marks

Identify the error in the following snippet.

```
int binarySearch(int arr[], int n, int key) {
    int low = 0, high = n - 1;
    while(low < high) {
        int mid = (low + high) / 2;
        if(arr[mid] == key)
            return mid;
        else if(arr[mid] < key)
            low = mid;
        else
            high = mid;
    }
    return -1;
}
```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>
2
3 int binarySearch(int arr[], int n, int key)
4 {
5     int low = 0, high = n - 1;
6
7     while (low <= high)
8     {
9         int mid = (low + high) / 2;
10
11         if (arr[mid] == key)
12             return mid;
13         else if (arr[mid] < key)
14             low = mid + 1;
15         else
16             high = mid - 1;
17     }
18
19     return -1;
20 }
21
22 int main()
23 {
24     int n, key;
25     scanf("%d", &n);
26
27     int arr[n];
28     for (int i = 0; i < n; i++)
29         scanf("%d", &arr[i]);
30
31     scanf("%d", &key);
32
33     printf("%d", binarySearch(arr, n, key));
34
35     return 0;
36 }
```

Test Input

```
5
1 2 3 4 5
3
```

Output

CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1)

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QUESTIONS

Q1

Binary Search

5 marks



Q2

Stack

5 marks



Q3

Stack

5 marks



Q4

Stack

5 marks



Q5

Stack

5 marks



Q6

Linked List

5 marks



Q7

Stack

5 marks



Q8

Linked List Data Structures

5 marks



Q9

Linked List

5 marks



Q10

Queue

5 marks



10/10 answered

Q2 Stack

5 marks

Find the error in the stack deletion operation. Justify it.

```
int pop() {  
    if(top == 0) {  
        printf("Underflow");  
        return -1;  
    }  
    return stack[top--];  
}
```

✳ Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>  
2  
3 int stack[100];  
4 int top = -1;  
5  
6 int pop()  
7 {  
8     if (top == -1)  
9     {  
10         printf("Underflow\n");  
11         return -1;  
12     }  
13     return stack[top--];  
14 }  
15  
16 int main()  
17 {  
18     pop();  
19     return 0;  
20 }
```

Test Input

stdin input (optional)

Output

Underflow

🕒 Pending manual review

✅ Submitted

CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1)

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QUESTIONS

Q1
Binary Search
5 marks**Q2**
Stack
5 marks**Q3**
Stack
5 marks**Q4**
Stack
5 marks**Q5**
Stack
5 marks**Q6**
Linked List
5 marks**Q7**
Stack
5 marks**Q8**
Linked List Data Structures
5 marks**Q9**
Linked List
5 marks**Q10**
Queue
5 marks

10/10 answered

Q3 Stack

5 marks

Check the snippet and identify the logical error.

```
int stack[5];  
int top;  
void init() {  
    top = 0;  
}
```

✳ Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>  
2  
3 int stack[5];  
4 int top;  
5  
6 void init()  
7 {  
8     top = -1;  
9 }  
10  
11 int main()  
12 {  
13     init();  
14     printf("%d", top);  
15     return 0;  
16 }
```

Test Input

stdin input (optional)

Output

-1

Pending manual review

Submitted

CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1)

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QUESTIONS

Q1
Binary Search
5 marks**Q2**
Stack
5 marks**Q3**
Stack
5 marks**Q4**
Stack
5 marks**Q5**
Stack
5 marks**Q6**
Linked List
5 marks**Q7**
Stack
5 marks**Q8**
Linked List Data Structures
5 marks**Q9**
Linked List
5 marks**Q10**
Queue
5 marks

10/10 answered

Q4 Stack

5 marks

Identify the errors in the following snippet.

```
if(ch == ')') {  
    while(stack[top] != ')') {  
        postfix[j++] = pop();  
    }  
    pop();  
}
```

✳ **Debugging Task:** Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>  
2  
3 char stack[100];  
4 int top = 0;  
5  
6 char pop()  
7 {  
8     return stack[top--];  
9 }  
10  
11 int main()  
12 {  
13     char ch = ')';  
14     if (ch == ')')  
15     {  
16         while (top != -1 && stack[top] != '(')  
17         {  
18             pop();  
19         }  
20  
21         if (top != -1)  
22             pop();  
23     }  
24  
25     printf("No Error");  
26     return 0;  
27 }
```

Test Input

stdin input (optional)

Output

No Error

Pending manual review

Submitted

CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1)

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QUESTIONS

Q1
Binary Search
5 marks**Q2**
Stack
5 marks**Q3**
Stack
5 marks**Q4**
Stack
5 marks**Q5**
Stack
5 marks**Q6**
Linked List
5 marks**Q7**
Stack
5 marks**Q8**
Linked List Data Structures
5 marks**Q9**
Linked List
5 marks**Q10**
Queue
5 marks

10/10 answered

Q5 Stack

5 marks

In balancing Bracket, you identify the error.

```
if((ch == '[' && stack[top] == '[') ||
(ch == ']' && stack[top] == '[')) {
    pop();
} else {
    return 0;
}
```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>
2
3 char stack[100];
4 int top = 0;
5
6 char pop()
7 {
8     return stack[top--];
9 }
10
11 int main()
12 {
13     char ch = '[';
14     stack[++top] = '[';
15
16     if (top != -1 &&
17         ((ch == '[' && stack[top] == '[') ||
18          (ch == ']' && stack[top] == '[') ||
19          (ch == '[' && stack[top] == '(')))
20     {
21         pop();
22         printf("Balanced");
23     }
24     else
25     {
26         printf("Not Balanced");
27     }
28 }
29
30 return 0;
31 }
```

Test Input

stdin input (optional)

Output

Balanced

⌚ Pending manual review

✔ Submitted

CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1)

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QUESTIONS

Q1 ✓
Binary Search
5 marks

Q2 ✓
Stack
5 marks

Q3 ✓
Stack
5 marks

Q4 ✓
Stack
5 marks

Q5 ✓
Stack
5 marks

Q6 ✓
Linked List
5 marks

Q7 ✓
Stack
5 marks

Q8 ✓
Linked List Data Structures
5 marks

Q9 ✓
Linked List
5 marks

Q10 ✓
Queue
5 marks

10/10 answered

Q6 Linked List

5 marks

What is the issue?
temp = head
while temp.next != NULL:
print(temp.data)
temp = temp.next

🔧 Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 temp = head
2 while temp != NULL:
3     print(temp.data)
4     temp = temp.next
```

Test Input

stdin input (optional)

Output

 Run Save Submit Fix

CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1)

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QUESTIONS

Q1

Binary Search
5 marks

Q2

Stack
5 marks

Q3

Stack
5 marks

Q4

Stack
5 marks

Q5

Stack
5 marks

Q6

Linked List
5 marks

Q7

Stack
5 marks

Q8

Linked List Data Structures
5 marks

Q9

Linked List
5 marks

Q10

Queue
5 marks

10/10 answered

Q8 Linked List Data Structures

5 marks

Identify the errors in the following code

while current != NULL:

current.next = prev

prev = current

current = current.next

🔧 Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>
2 #include <stdlib.h>
3
4 struct Node {
5     int data;
6     struct Node *next;
7 };
8
9 int main() {
10     struct Node *prev = NULL;
11     struct Node *current;
12     struct Node *next;
13
14     /* Create nodes */
15     struct Node *head = (struct Node *)malloc(sizeof(struct Node));
16     struct Node *second = (struct Node *)malloc(sizeof(struct Node));
17     struct Node *third = (struct Node *)malloc(sizeof(struct Node));
18
19     head->data = 10;
20     head->next = second;
21
22     second->data = 20;
23     second->next = third;
24
25     third->data = 30;
26     third->next = NULL;
27
28     current = head;
29
30     /* Reverse the linked list */
31     while (current != NULL) {
32         next = current->next;
33         current->next = prev;
34         prev = current;
35         current = next;
36     }
37
38     head = prev;
39
40     /* Display reversed list */
41     current = head;
42
43     while (current != NULL) {
44         printf("%d ", current->data);
45         current = current->next;
46     }
47
48     return 0;
49 }
```

Test Input

Output

```
30 20 10
```




CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1)

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QUESTIONS

Q1

Binary Search
5 marks



Q2

Stack
5 marks



Q3

Stack
5 marks



Q4

Stack
5 marks



Q5

Stack
5 marks



Q6

Linked List
5 marks



Q7

Stack
5 marks



Q8

Linked List Data Structures
5 marks



Q9

Linked List
5 marks



Q10

Queue
5 marks



10/10 answered

Q9 Linked List

5 marks

What is the error in the below code?

```
new.next = head
```

```
head = new
```

🔧 Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 new->next = head;  
2 head = new;
```

Test Input

stdin input (optional)

Output



Run



Save



Submit Fix

CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1)

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QUESTIONS

Q1 ✓
Binary Search
5 marks

Q2 ✓
Stack
5 marks

Q3 ✓
Stack
5 marks

Q4 ✓
Stack
5 marks

Q5 ✓
Stack
5 marks

Q6 ✓
Linked List
5 marks

Q7 ✓
Stack
5 marks

Q8 ✓
Linked List Data Structures
5 marks

Q9 ✓
Linked List
5 marks

Q10 ✓
Queue
5 marks

10/10 answered

Q10 Queue

5 marks

What is missing in the below pseudocode?

DEQUEUE(Q):

if front == -1:

print("Underflow")

else:

return Q[front++]

 **Debugging Task:** Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 if (front == -1 || front > rear) {
2     printf("Underflow");
3 } else {
4     return Q[front
```

Test Input

stdin input (optional)

Output

 Run

 Save

 Submit Fix

CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 2)

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QUESTIONS

Q1 ✓
Stack Notations - Postfix Expression
5 marks**Q2** ✓
Circular Queue
5 marks**Q3** ✓
Binary Search
5 marks**Q4** ✓
Queue
5 marks**Q5** ✓
Queue
5 marks**Q6** ✓
Doubly Linked List
5 marks**Q7** ✓
Stack
5 marks**Q8** ✓
Stack
5 marks**Q9** ✓
Circular Queue
5 marks**Q10** ✓
Stack
5 marks

10/10 answered

Q1 Stack Notations - Postfix Expression

5 marks

Find the bug in the evaluation of a Postfix expression.

```
while (expression[i] != '\0') {  
    if (isdigit(expression[i])) push(expression[i] - '0');  
    else {  
        int op1 = pop();  
        int op2 = pop();  
        // perform operation: result = op1 + op2;  
        push(result);  
    }  
}
```

🔧 **Debugging Task:** Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>  
2 #include <ctype.h>  
3  
4 int stack[100];  
5 int top = -1;  
6  
7 void push(int x)  
8 {  
9     stack[++top] = x;  
10 }  
11  
12 int pop()  
13 {  
14     return stack[top--];  
15 }  
16  
17 int main()  
18 {  
19     char expression[] = "23+";  
20     int i = 0;  
21     int op1, op2, result;  
22  
23     while (expression[i] != '\0')  
24     {  
25         if (isdigit(expression[i]))  
26         {  
27             push(expression[i] - '0');  
28         }  
29         else  
30         {  
31             op1 = pop();  
32             op2 = pop();  
33  
34             result = op2 + op1;  
35             push(result);  
36         }  
37  
38         i++;  
39     }  
40  
41     printf("%d", pop());  
42  
43     return 0;  
44 }
```

Test Input

23+

Output

5

CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 2)

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QUESTIONS

Q1 ✓
Stack Notations - Postfix Expression
5 marks

Q2 ✓
Circular Queue
5 marks

Q3 ✓
Binary Search
5 marks

Q4 ✓
Queue
5 marks

Q5 ✓
Queue
5 marks

Q6 ✓
Doubly Linked List
5 marks

Q7 ✓
Stack
5 marks

Q8 ✓
Stack
5 marks

Q9 ✓
Circular Queue
5 marks

Q10 ✓
Stack
5 marks

10/10 answered

Q3 Binary Search

5 marks

Identify the bug in this Recursive Binary Search.

```
int binarySearch(int arr[], int l, int r, int x) {
    int mid = l + (r - l) / 2;
    if (arr[mid] == x) return mid;
    if (arr[mid] > x) return binarySearch(arr, l, mid, x);
    return binarySearch(arr, mid, r, x);
}
```

 **Debugging Task:** Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 int binarySearch(int arr[], int l, int r, int x) {
2     if (l > r)
3         return -1;
4
5     int mid = l + (r - l) / 2;
6
7     if (arr[mid] == x)
8         return mid;
9
10    if (arr[mid] > x)
11        return binarySearch(arr, l, mid - 1, x);
12
13    return binarySearch(arr, mid + 1, r, x);
14 }
```

Test Input

stdin input (optional)

Output

 Pending manual review

✓ Submitted

CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 2)

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QUESTIONS

Q1

Stack Notations - Postfix
Expression
5 marks

Q2

Circular Queue
5 marks

Q3

Binary Search
5 marks

Q4

Queue
5 marks

Q5

Queue
5 marks

Q6

Doubly Linked List
5 marks

Q7

Stack
5 marks

Q8

Stack
5 marks

Q9

Circular Queue
5 marks

Q10

Stack
5 marks

10/10 answered

Q4 Queue

5 marks

Find the error in this Deque insertFront operation.

```
void insertFront(int data) {  
    if (front == 0) front = MAX - 1;  
    else front = front - 1;  
    deque[front] = data;  
}
```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 void insertFront(int data)  
2 {  
3     if ((front == 0 && rear == MAX - 1) || (front == rear + 1))  
4         return;  
5  
6     if (front == -1)  
7     {  
8         front = 0;  
9         rear = 0;  
10    }  
11    else if (front == 0)  
12    {  
13        front = MAX - 1;  
14    }  
15    else  
16    {  
17        front--;  
18    }  
19  
20    deque[front] = data;  
21 }
```

Test Input

stdin input (optional)

Output



Run



Save



Submit Fix

CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 2)

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QUESTIONS

Q1

Stack Notations - Postfix Expression
5 marks

Q2

Circular Queue
5 marks

Q3

Binary Search
5 marks

Q4

Queue
5 marks

Q5

Queue
5 marks

Q6

Doubly Linked List
5 marks

Q7

Stack
5 marks

Q8

Stack
5 marks

Q9

Circular Queue
5 marks

Q10

Stack
5 marks

10/10 answered

Q5 Queue

5 marks

Identify the error in the Queue enqueue logic.

```
void enqueue(int data) {
    if (rear == MAX) return;
    queue[rear++] = data;
}
```

🔧 Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 void enqueue(int data) {
2     if (rear == MAX - 1)
3         return;
4
5     queue[++rear] = data;
6 }
```

Test Input

stdin input (optional)

Output

```
Compilation Error:
C:\Windows\TEMP\sandbox_192521202RB_6a7c2b99cac8b5.57484365\main.c: In
function 'enqueue':
C:\Windows\TEMP\sandbox_192521202RB_6a7c2b99cac8b5.57484365\main.c:2:9:
error: 'rear' undeclared (first use in this function)
2 |     if (rear == MAX - 1)
  |     ~~~~~^
```

⌚ Pending manual review

✅ Submitted

CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 2)

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QUESTIONS

Q1
Stack Notations - Postfix Expression
5 marks

Q2
Circular Queue
5 marks

Q3
Binary Search
5 marks

Q4
Queue
5 marks

Q5
Queue
5 marks

Q6
Doubly Linked List
5 marks

Q7
Stack
5 marks

Q8
Stack
5 marks

Q9
Circular Queue
5 marks

Q10
Stack
5 marks

10/10 answered

Q6 Doubly Linked List

5 marks

Identify the error in this Doubly Linked List node deletion.

```
void deleteNode(Node* del) {
    if (del->prev != NULL)
        del->prev->next = del->next;
    if (del->next != NULL)
        del->next->prev = del->prev;
}
```

🔧 Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 void deleteNode(Node *del)
2 {
3     if (del == NULL)
4         return;
5
6     if (del->prev != NULL)
7         del->prev->next = del->next;
8     else
9         head = del->next;
10
11     if (del->next != NULL)
12         del->next->prev = del->prev;
13 }
```

Test Input

stdin input (optional)

Output

```
Compilation Error:
C:\Windows\TEMP\sandbox_192521202RB_6a7c2be8df6e54.64098397\main.c:1:17:
error: unknown type name 'Node'
1 | void deleteNode(Node *del)
  |                  ^~~~~
```

 Pending manual review

 Submitted

CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 2)

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QUESTIONS

Q1 ✓
Stack Notations - Postfix Expression
5 marks

Q2 ✓
Circular Queue
5 marks

Q3 ✓
Binary Search
5 marks

Q4 ✓
Queue
5 marks

Q5 ✓
Queue
5 marks

Q6 ✓
Doubly Linked List
5 marks

Q7 ✓
Stack
5 marks

Q8 ✓
Stack
5 marks

Q9 ✓
Circular Queue
5 marks

Q10 ✓
Stack
5 marks

10/10 answered

Q7 Stack

5 marks

Identify the issue in the below pseudocode.

POP():

if top == 0 → Underflow

🔧 Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 int pop()
2 {
3     if (top == -1)
4         return -1;
5
6     return stack[top--];
7 }
```

Test Input

stdin input (optional)

Output

```
Compilation Error:
C:\Windows\TEMP\sandbox_192521202RB_6a7c2c1a2f4e07.76803801\main.c: In
function 'pop':
C:\Windows\TEMP\sandbox_192521202RB_6a7c2c1a2f4e07.76803801\main.c:3:9:
error: 'top' undeclared (first use in this function); did you mean 'pop'?
3     if (top == -1)
```

 Pending manual review

 Submitted

CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 2)

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QUESTIONS

Q1 ✓
Stack Notations - Postfix Expression
5 marks

Q2 ✓
Circular Queue
5 marks

Q3 ✓
Binary Search
5 marks

Q4 ✓
Queue
5 marks

Q5 ✓
Queue
5 marks

Q6 ✓
Doubly Linked List
5 marks

Q7 ✓
Stack
5 marks

Q8 ✓
Stack
5 marks

Q9 ✓
Circular Queue
5 marks

Q10 ✓
Stack
5 marks

10/10 answered

Q9 Circular Queue

5 marks

Identify the issue in the following snippet for Circular Queue in Traffic Signal.

```
rear = (rear + 1) % MAX;
queue[rear] = vehicle;
if(rear == front){
printf("Queue Full");
}
```

🔧 Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 if ((rear + 1) % MAX == front)
2 {
3     printf("Queue Full");
4 }
5 else
6 {
7     rear = (rear + 1) % MAX;
8     queue[rear] = vehicle;
9 }
```

Test Input

stdin input (optional)

Output

 Pending manual review

 Submitted